

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456971.7282 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.0098
Transit depth (Rp/Rs)^2	1.3 +/- 0.22 [%]
Orbital Inclination (inc)	89.85 +/- 1.1
Airmass coefficient 1 (a1)	1.0178 +/- 0.0069
Airmass coefficient 2 (a2)	-0.0152 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	#1 - [211, 99]
Optimal Aperture	3.74
Optimal Annulus	14.02
Transit Duration (day)	0.1553 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.114 ± 0.0098 vs 0.1036 (deviation 1.06σ)
Duration fit vs published	0.1553 d vs 0.1567 d (deviation 0.8σ)
Mid-time O–C	-2.5 min
Depth SNR	11.6
Residual scatter	0.68 %

Light curve

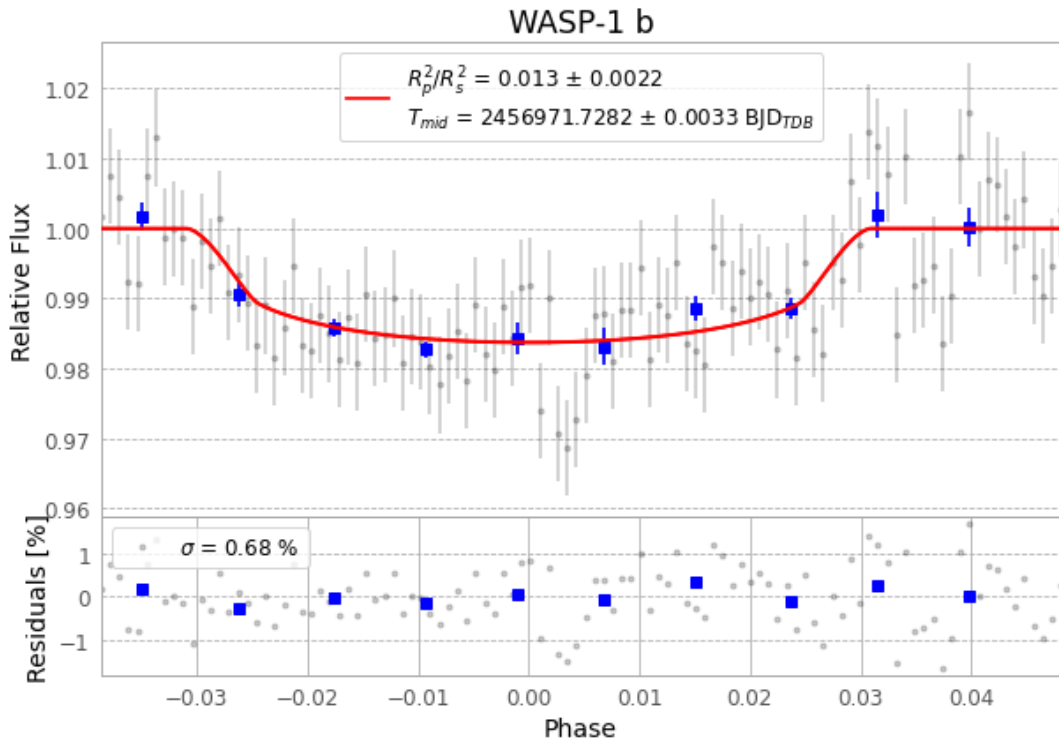


Figure 1 — FinalLightCurve_WASP-1 b_2014-11-09.png (SHA-256 6a7454a197b63d6a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [485, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7b3f20693dd0d054...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

106 FITS frames (70 MB), WASP-1141110030012.FITS → WASP-1141110081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-141110.log (14831aa01dfa...), FinalLightCurve_WASP-1 b_2014-11-09.png (6a7454a197b6...), FinalLightCurve_WASP-1 b_2014-11-09.csv (3d0da3520c53...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:02Z.